project proposal

1. Set up

Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)

library(dplyr)     # data manipulation
library(broom)     # tidy statistical output
library(corrplot)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")


# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

2. Cleaning

`sites` object

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

Cleaning `taxon_list`

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

Cleaning water_quality

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

Cleaning aq_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
```

```
        diptera_chironomid,
        annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

Focus Taxa

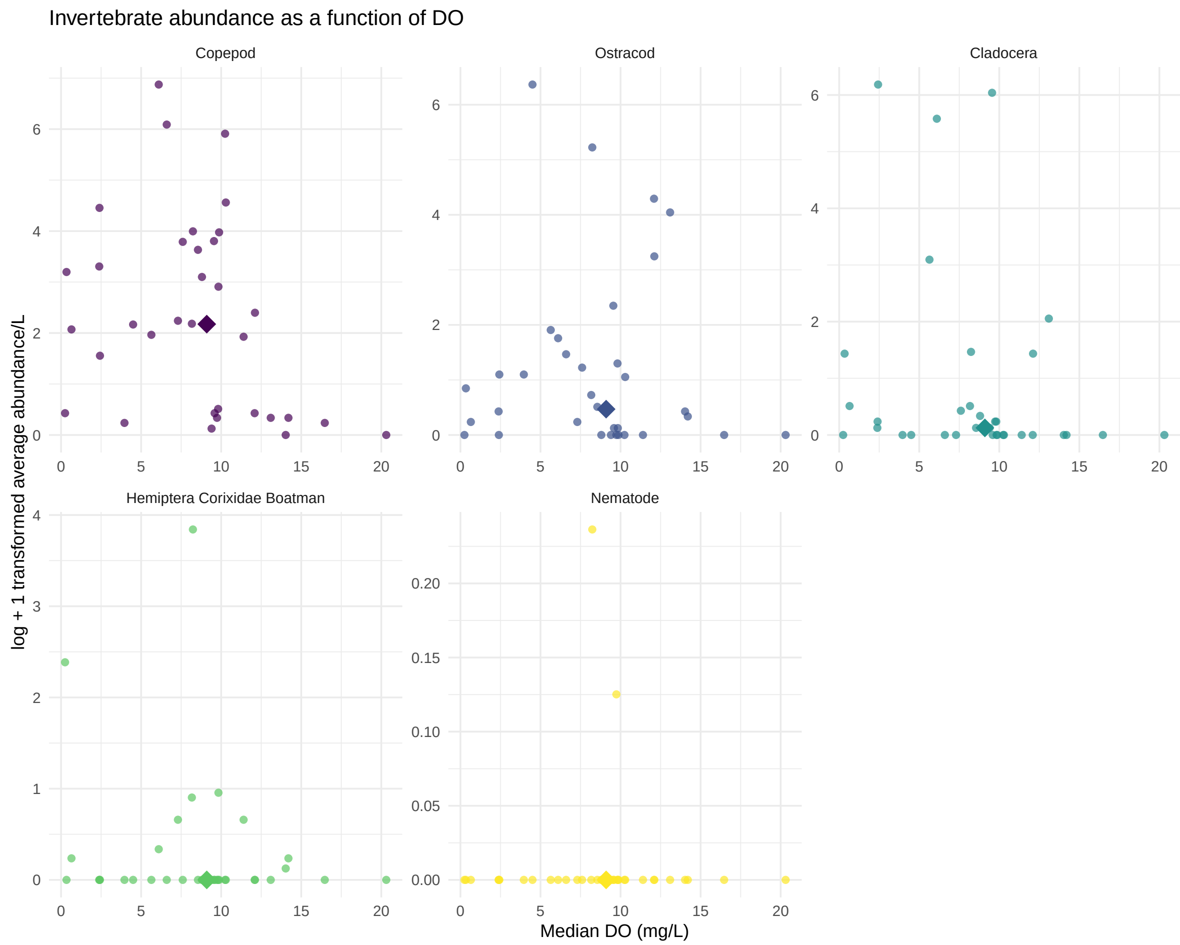
```
# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")
```

Figure 1: Invertebrate abundance as a function of DO

```r
# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)

# creates median point location for each taxon
DO_taxa_medians <- DO_taxa |>
  # groups data by taxon so each species gets one median point
  group_by(taxon) |>
  # calculates median DO and median log abundance for each taxon
  summarise(
    med_do_point = median(med_do, na.rm = TRUE),
    med_log_ave_lit = median(log_ave_lit, na.rm = TRUE),
    .groups = "drop"
)
```

```r
DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
                        # uses DO_taxa data frame
                        mapping = aes(x = med_do, # x-axis: med_do
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7

  # second layer: larger point shows the median location for each taxon
  geom_point(data = DO_taxa_medians,
             mapping = aes(x = med_do_point,
```

```
                            y = med_log_ave_lit),
              shape = 18,
              size = 5) +

  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "Median DO (mg/L)", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance as a function of DO") +

  # uses non-default theme
  theme_minimal() +
  # remove redundant legend
  theme(legend.position = "none")
DO_taxa_plot
```



shows that individual taxa have noisy, weak, or inconsistent relationships with DO.

Figure 2: Variance of DO across sights

```
# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <-  aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include focus taxa
  filter(taxon %in% focus_taxa)

# base layer
```

```r
DO <- ggplot(data = aq_ins_clean_fig2,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_do)) +
  # first layer: boxplot
  geom_boxplot(width = 0.35,
               outlier.shape = NA) +
  # second layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.12,
              shape = 21,
              alpha = 0.6) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median DO (mh/L)",
       title = "Site DO levels") +
  # different theme
  theme_bw()
# show plot
DO
```

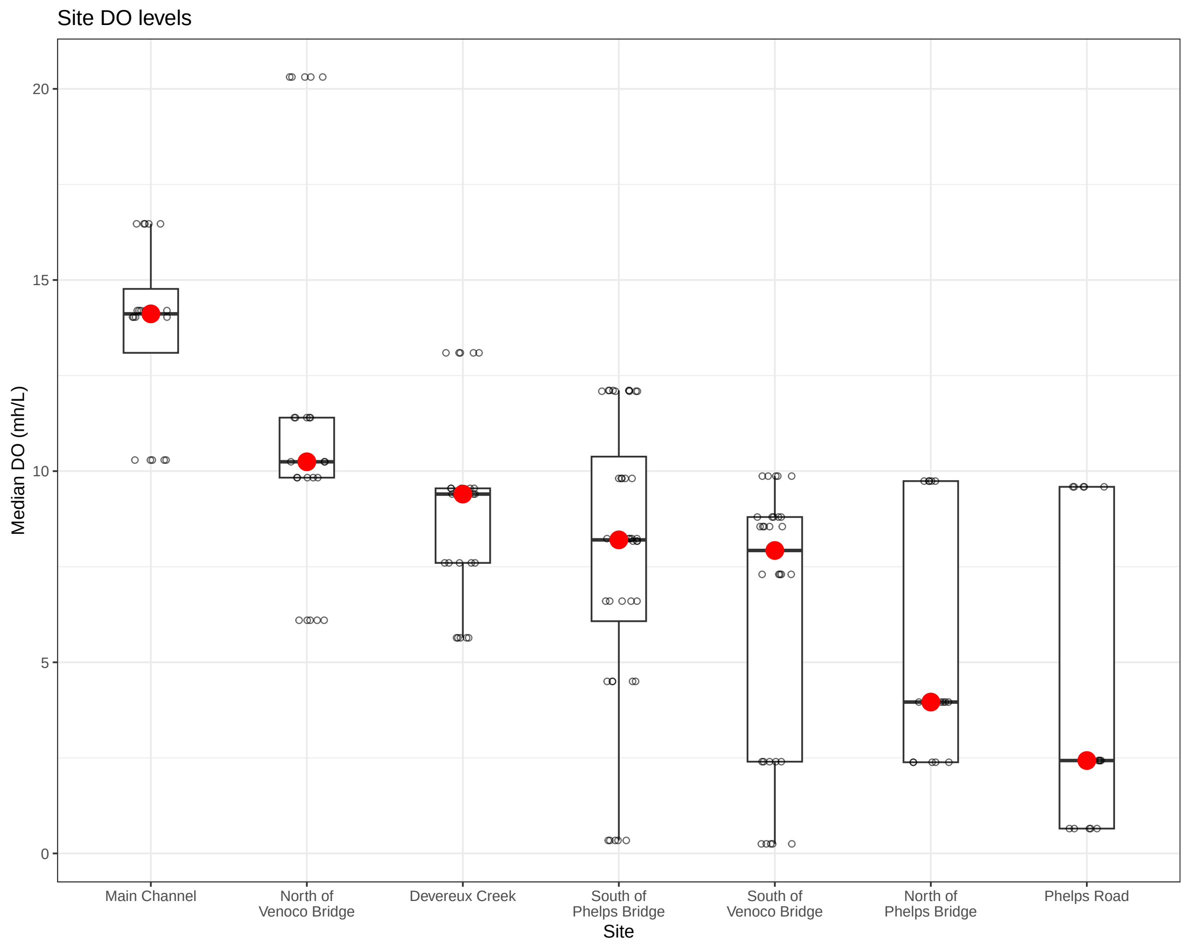


Figure 3: Aquatic Invert. Abundance Across Sites

```
# creating new object from clean aquatic inverts
abundance_by_site <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include taxa of interest
  filter(taxon %in% focus_taxa)
abundance <- ggplot(data = abundance_by_site,
                   # x-axis: site
```

```
                   # y-axis: mean abundance/L
                   # color by taxon
                   mapping = aes(x = site_full,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # first layer: jitter points to represent observations
  geom_jitter(width = 0.15,
              height = 0,
              alpha = 0.7) +
  # facet by taxon
  facet_wrap(~ taxon) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # non-default colors
  scale_color_viridis_d() +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Aquatic Invert. Abundance Across Sites") +
  theme_bw() +
  # remove legend
  theme(legend.position = 'none',
        # fix x-axis for readability
        axis.text.x = element_text(angle = 45, hjust = 1))
```

```
# base layer

abundance <- ggplot(data = abundance_by_site,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   # color by taxon
                   mapping = aes(x = site_full,
                                 y = log_ave_lit,
                                 color = taxon)) +
  # first layer: jitter points to represent observations
  geom_jitter(width = 0.15,
              height = 0,
              alpha = 0.7) +
  # facet by taxon
  facet_wrap(~ taxon) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
```

```
  # non-default colors
  scale_color_viridis_d() +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Aquatic Invert. Abundance Across Sites") +
  theme_bw() +
  # remove legend
  theme(legend.position = 'none',
        # fix x-axis for readability
        axis.text.x = element_text(angle = 45, hjust = 1))

abundance
```

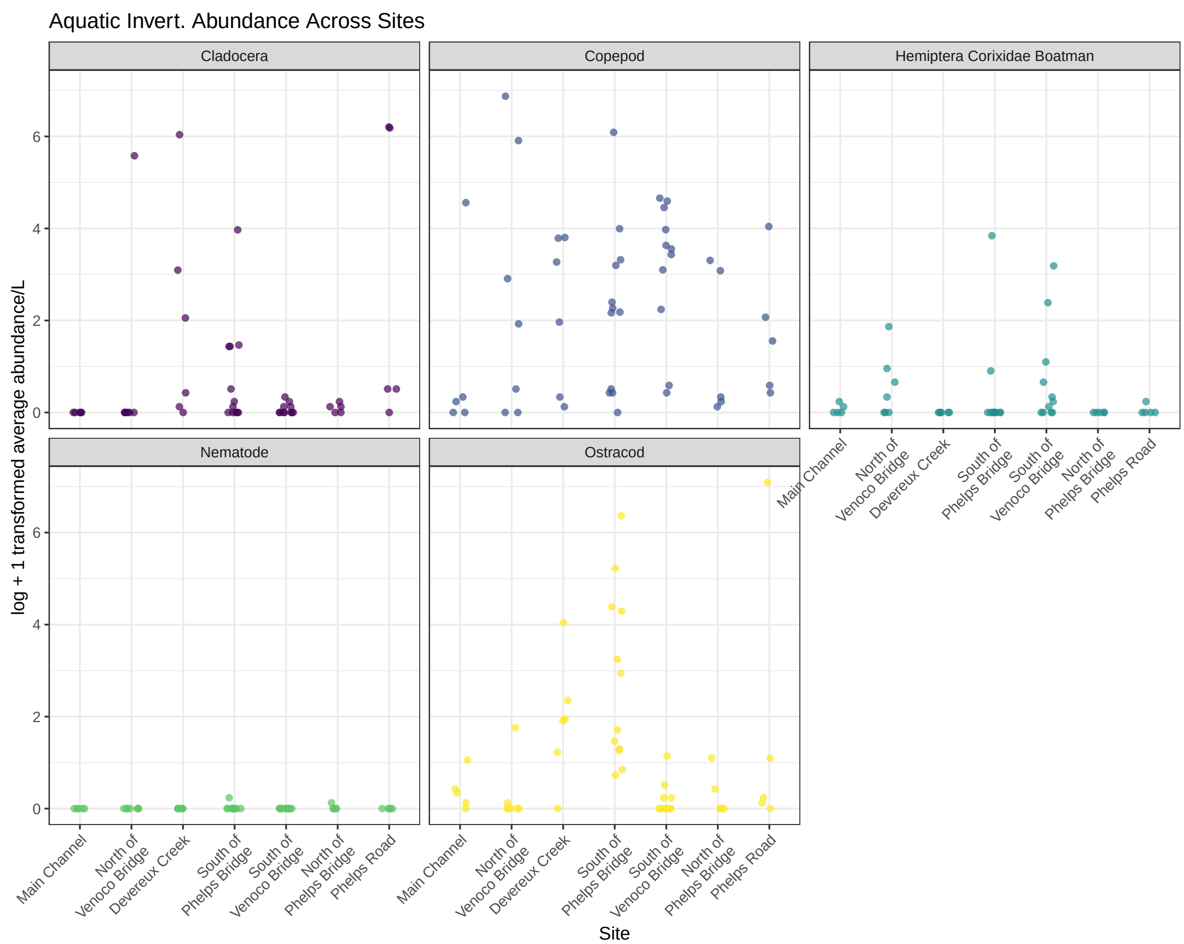
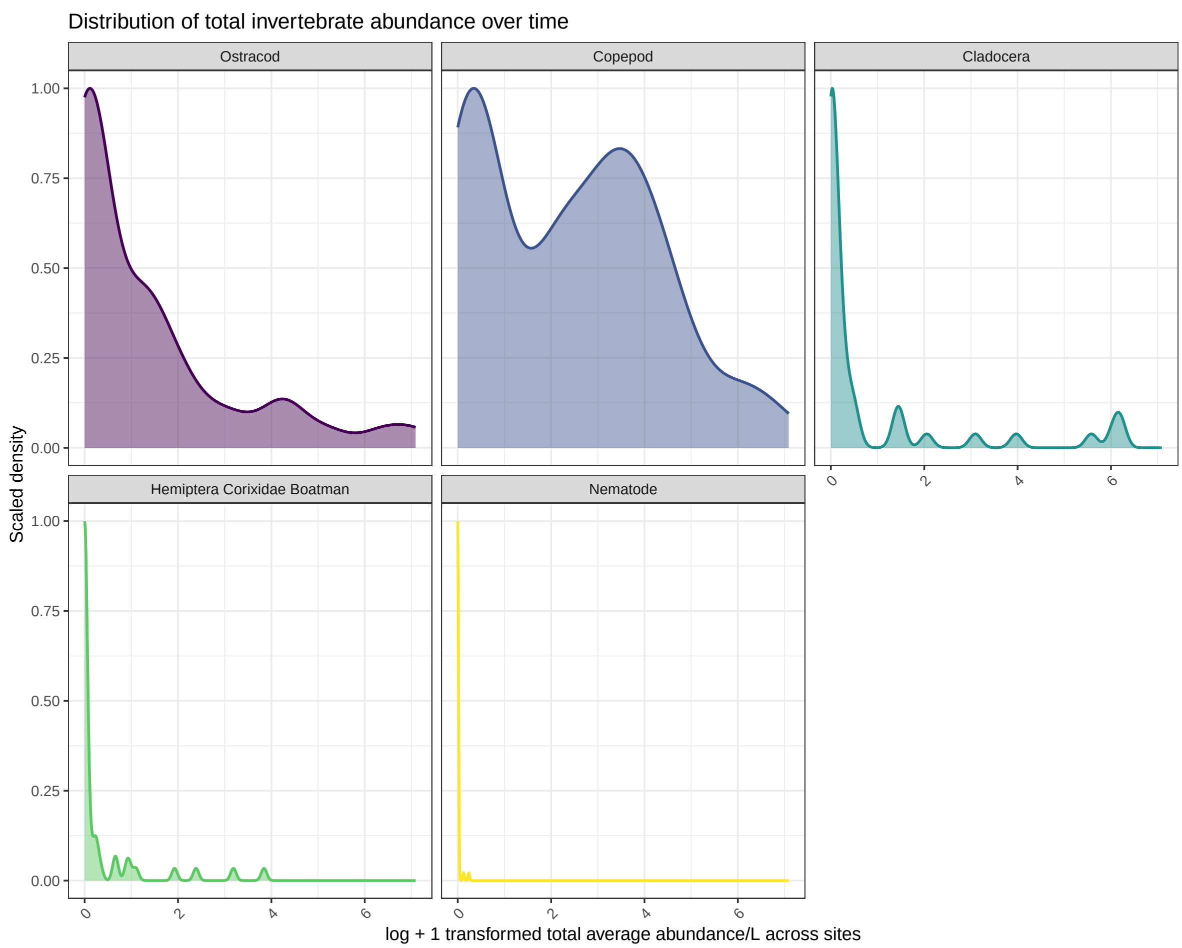


Figure 4: Abundance over time

```r
# create object for plotting total abundance over time
abundance_time <- aq_ins_clean |>
  # group by sampling date and taxon
  group_by(date, taxon) |>
  # total abundance across all sites for each date and taxon
  summarize(total_ave_lit = sum(ave_lit, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create log-transformed total abundance column
  mutate(log_total_ave_lit = log(total_ave_lit + 1)) |>
  # make taxon names easier to read
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # order taxa by maximum total abundance
  mutate(taxon = fct_reorder(taxon, total_ave_lit, .fun = max, .desc = TRUE)) |>
  # include main taxon groups of focus
  filter(taxon %in% focus_taxa)


# base layer
abundance_time_plot <- ggplot(data = abundance_time,
                              # x-axis: log_total_ave_lit
                              # fill and color by taxon
                              mapping = aes(x = log_total_ave_lit,
                                            fill = taxon,
                                            color = taxon)) +
  # scaled density makes y-axis consistent across facets
  geom_density(aes(y = after_stat(scaled)),
               alpha = 0.45,
               linewidth = 0.8,
               adjust = 0.8) +
  facet_wrap(~ taxon) + # facet by taxon
  scale_fill_viridis_d() + # non-default colors
  scale_color_viridis_d() + # non-default colors
  labs(x = "log + 1 transformed total average abundance/L across sites",
       y = "Scaled density",
       title = "Distribution of total invertebrate abundance over time") +
  # use clean theme
  theme_bw() +
```

```r
  # remove redundant legend
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45,
                                   hjust = 1))

# show plot
abundance_time_plot
```

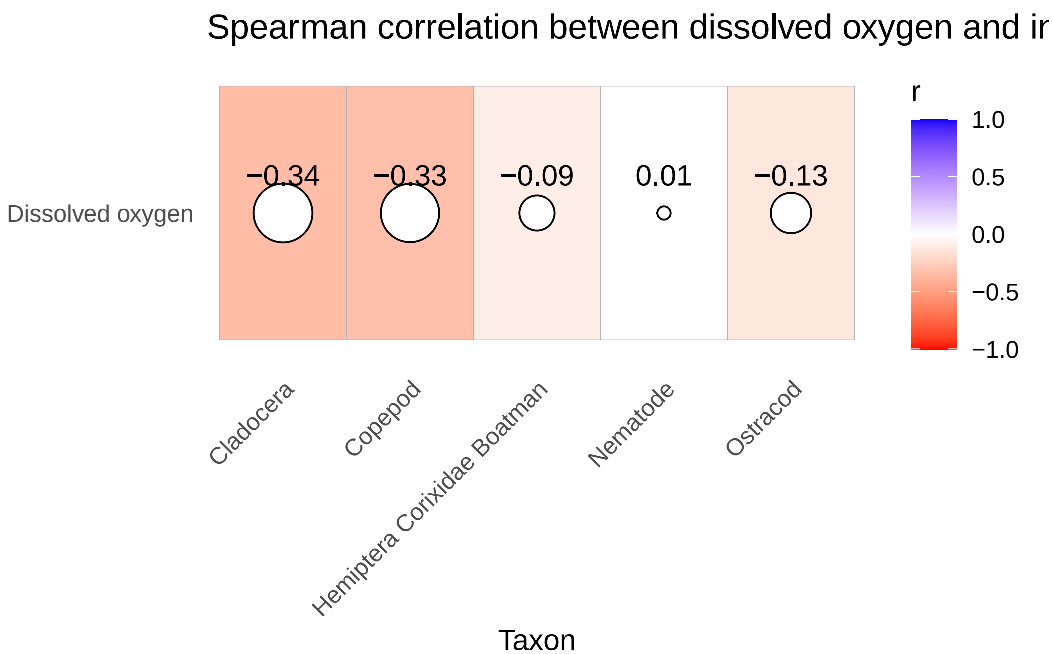


Correlation analysis

```r
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
```

```r
  select(date, site, taxon, log_ave_lit, med_do) |>
  pivot_wider(
    names_from = taxon,
    values_from = log_ave_lit
  )

# calculate correlation of each taxon with dissolved oxygen
# calculate Spearman correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  select(-date, -site) |>
  summarize(across(-med_do, ~ cor(.x,
                                  med_do,
                                  method = "spearman",
                                  use = "complete.obs"))) |>
  pivot_longer(cols = everything(),
               names_to = "taxon",
               values_to = "correlation")

ggplot(cor_do, aes(x = taxon, y = "Dissolved oxygen", fill = correlation)) +
  geom_tile(color = "grey70") +
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +

  # move labels ABOVE circles
  geom_text(aes(label = round(correlation, 2)),
            nudge_y = 0.15,  # <-- this moves text up
            size = 4) +

  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0,
    limits = c(-1, 1),
    name = "r"
  ) +
  scale_size(range = c(2, 10), guide = "none") +
  labs(
    x = "Taxon",
    y = NULL,
    title = "Spearman correlation between dissolved oxygen and invertebrate abundance"
  ) +
  theme_minimal() +
```

```
  theme(
    panel.grid = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```




```
# Spearman correlation between DO and abundance for each taxon
spearman_by_taxon <- DO_taxa |>
  # keep only species of interest
  filter(taxon %in% focus_taxa) |>
  # group by taxon so each species gets its own test
  group_by(taxon) |>
  # run Spearman correlation test for each taxon
  summarise(
    spearman_rho = cor(log_ave_lit,
                       med_do,
                       method = "spearman",
                       use = "complete.obs"),
    p_value = cor.test(log_ave_lit,
                       med_do,
                       method = "spearman",
                       exact = FALSE)$p.value,
    n = sum(complete.cases(log_ave_lit, med_do)),
```

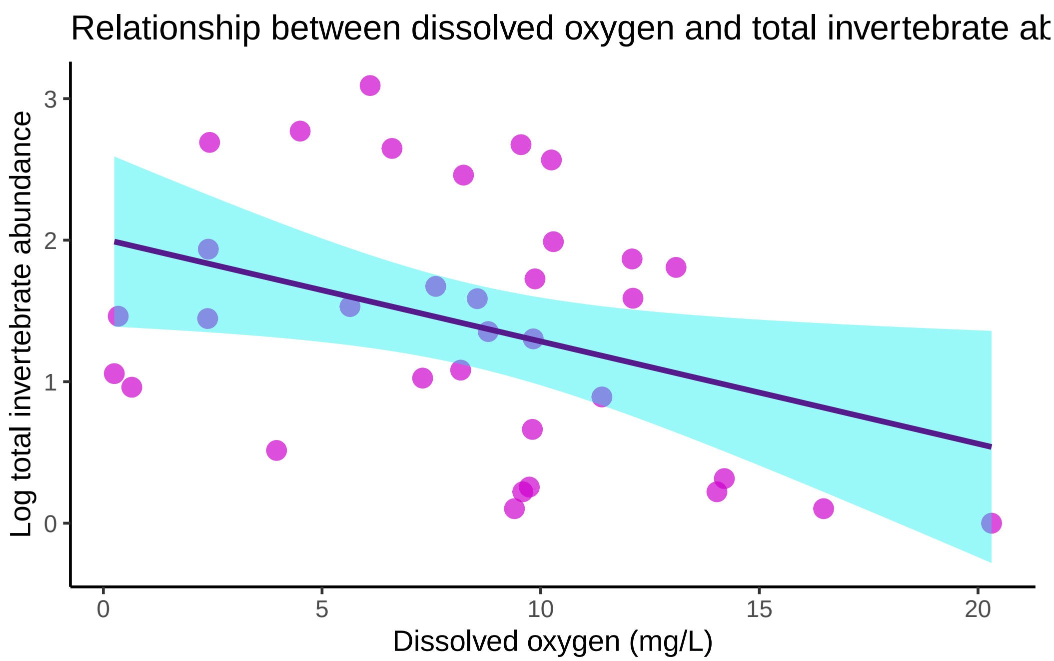
```
    .groups = "drop"
  )

# show results
spearman_by_taxon
```

```
# A tibble: 5 x 4
  taxon                       spearman_rho p_value     n
  <fct>                              <dbl>   <dbl> <int>
1 Copepod                          -0.335   0.0529    34
2 Ostracod                         -0.128   0.470     34
3 Cladocera                        -0.342   0.0475    34
4 Hemiptera Corixidae Boatman      -0.0869  0.625     34
5 Nematode                          0.0105  0.953     34
```

Summed abundance across focus taxa vs DO

```
focus_taxa <- c("Cladocera", "Copepod",
                "Hemiptera Corixidae Boatman",
                "Nematode", "Ostracod")

total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    log_total_abundance = log10(total_abundance + 1),
    .groups = "drop"
  )
ggplot(total_abundance,
       aes(x = med_do,
           y = log_total_abundance)) +

  geom_point(color = "magenta3",
             size = 3,
             alpha = 0.7) +
   geom_smooth(method = "lm",
              se = TRUE,
              color = "purple4",
```

```
              fill = "cyan2") +

  labs(
    x = "Dissolved oxygen (mg/L)",
    y = "Log total invertebrate abundance",
    title = "Relationship between dissolved oxygen and total invertebrate abundance"
  ) +

  theme_classic()
```



When taxa are combined, a clearer negative relationship emerges.

```
cor.test(
  total_abundance$log_total_abundance,
  total_abundance$med_do,
  method = "spearman",
  exact = FALSE)
```

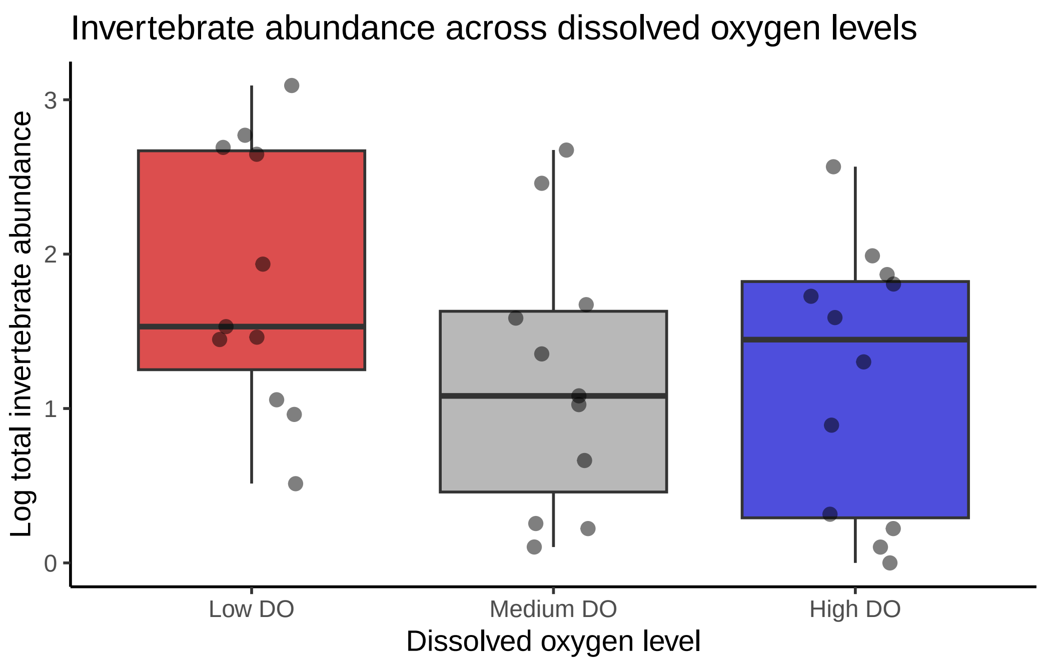
```
    Spearman's rank correlation rho

data:  total_abundance$log_total_abundance and total_abundance$med_do
```

```
S = 8526.3, p-value = 0.08182
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.3027201
```

Binned DO levels

```
total_abundance <- total_abundance |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    )
  )
total_abundance$do_category <- factor(
  total_abundance$do_category,
  levels = c("Low DO", "Medium DO", "High DO")
)

ggplot(total_abundance,
       aes(x = do_category,
           y = log_total_abundance,
           fill = do_category)) +

  geom_boxplot(alpha = 0.7,
               outlier.color = "black",
               outlier.size = 2) +

  # optional: add jittered points to show raw data
  geom_jitter(width = 0.15,
              alpha = 0.5,
              size = 2,
              color = "black") +

  labs(
    x = "Dissolved oxygen level",
    y = "Log total invertebrate abundance",
    title = "Invertebrate abundance across dissolved oxygen levels"
```

```r
  ) +

  scale_fill_manual(values = c(
    "Low DO" = "red3",
    "Medium DO" = "grey60",
    "High DO" = "blue3"
  )) +

  theme_classic() +
  theme(legend.position = "none")
```



Community abundance varies continuously rather than discretely across DO levels.

```r
# Pearson correlation for continuous DO and total community abundance
cor.test(total_abundance$log_total_abundance,
         total_abundance$med_do,
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  total_abundance$log_total_abundance and total_abundance$med_do
```

```
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
       cor
-0.3749894
```

```
# r = -0.3749894, p = 0.02887
# there is a significant negative correlation between abundance and continous DO

# Kruskal-Wallis test comparing abundance across DO categories
kruskal.test(log_total_abundance ~ do_category,
             data = total_abundance)
```

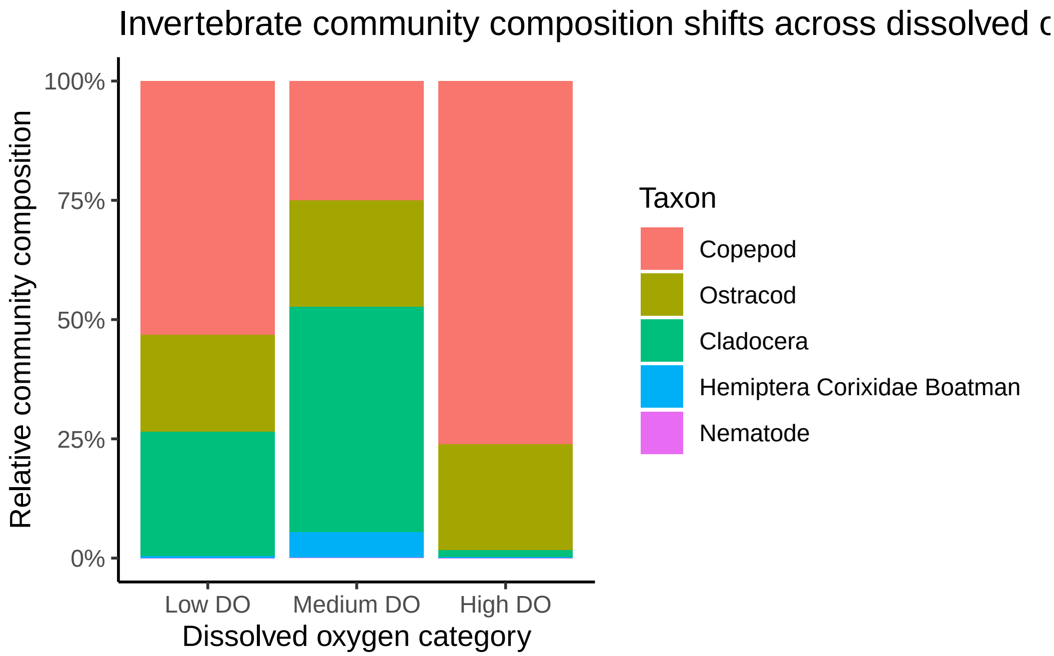
```

	Kruskal-Wallis rank sum test

data:  log_total_abundance by do_category
Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
```

```
# Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
# there is not a significant difference between abundance and DO categories
```

Composition plot across DO bins

```
composition_plot <- DO_taxa |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    ),

    # reorder factor levels
    do_category = factor(do_category,
                         levels = c("Low DO",
                                    "Medium DO",
                                    "High DO"))
```

```r
  ) |>
  group_by(do_category, taxon) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  ggplot(aes(x = do_category,
             y = total_abundance,
             fill = taxon)) +

  geom_col(position = "fill") +

  scale_y_continuous(labels = scales::percent_format()) +

  labs(
    x = "Dissolved oxygen category",
    y = "Relative community composition",
    fill = "Taxon",
    title = "Invertebrate community composition shifts across dissolved oxygen levels"
  ) +

  theme_classic()

composition_plot
```

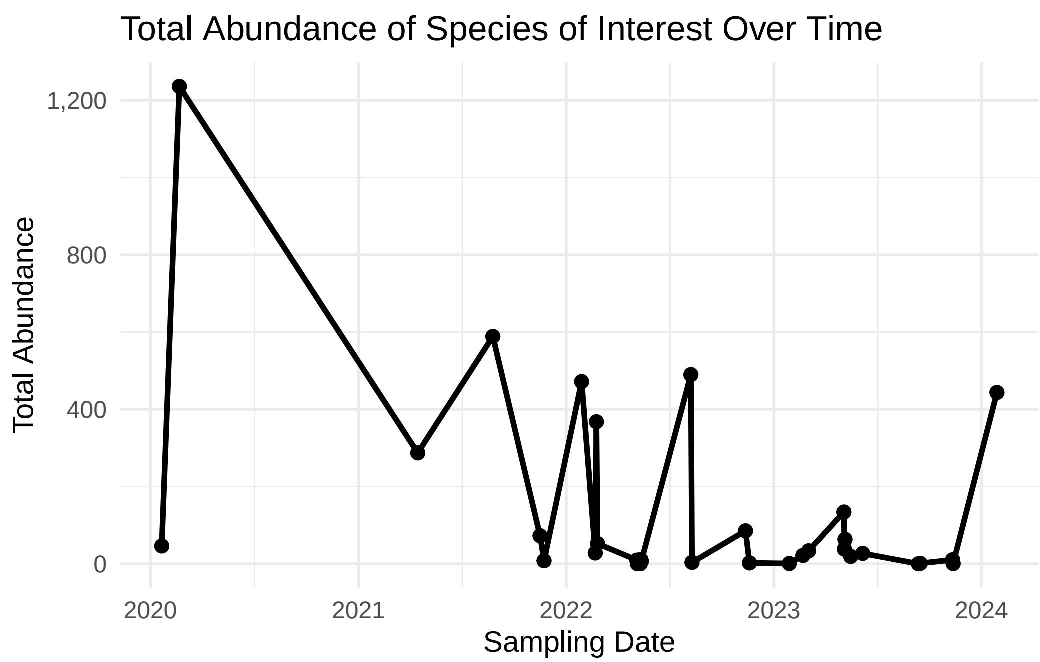


Medium DO sites show a more even distribution of taxa and greater relative abundance of Cladocera and Ostracods, while low and high DO conditions are dominated primarily by Copepods.

Time-series plot (all species)

```
# species/taxa to include in the total abundance figure
species_interest <- c(
  "Ostracod",
  "Copepod",
  "Cladocera",
  "Hemiptera Corixidae Boatman",
  "Nematode"
)

# create total abundance over time
total_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% species_interest) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
```

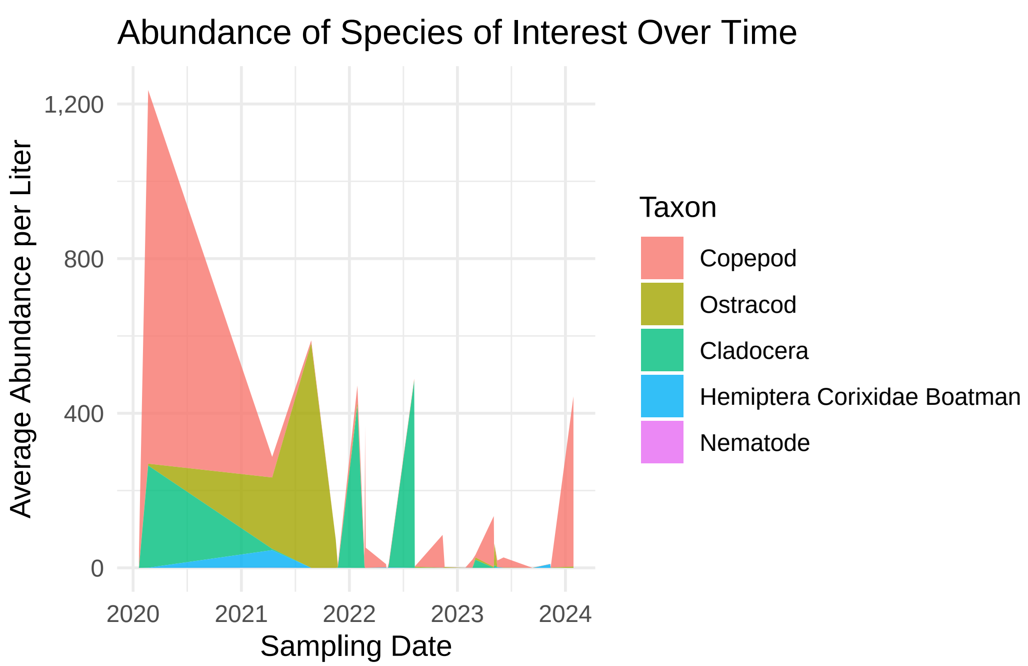
```r
  # groups by date only, so abundance is summed across sites and taxa
  group_by(date) |>
  # sums average abundance per liter across all taxa of interest
  summarise(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# plot total abundance over time
ggplot(total_abundance_time, aes(x = date, y = total_abundance)) +
  # adds a line connecting total abundance values over time
  geom_line(linewidth = 1) +
  # adds points for each sampling date
  geom_point(size = 2) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Total Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Total Abundance"
  ) +
  # uses a clean theme
  theme_minimal()
```



Time-series area plot

```
# species/taxa to include in the area plot
species_interest <- c(
  "Ostracod",
  "Copepod",
  "Cladocera",
  "Hemiptera Corixidae Boatman",
  "Nematode"
)

# create abundance over time by taxon
taxon_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% species_interest) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
  # groups by date and taxon so each species stays separate
  group_by(date, taxon) |>
  # sums across sites, but not across taxa
```

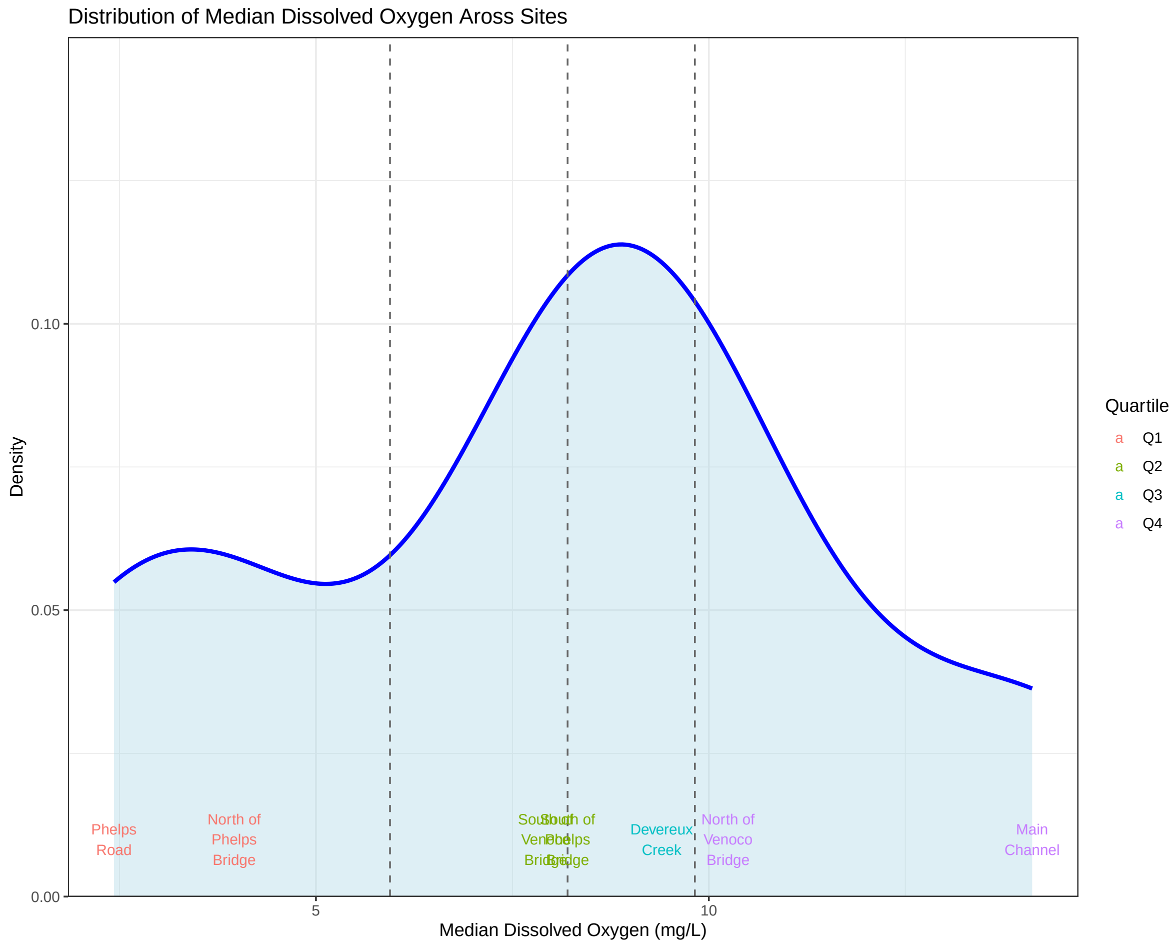
```r
  summarise(
    ave_lit = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# create time series area plot by taxon
ggplot(taxon_abundance_time, aes(x = date, y = ave_lit, fill = taxon)) +
  # creates stacked area plot with a different color for each taxon
  geom_area(alpha = 0.8) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Average Abundance per Liter",
    fill = "Taxon"
  ) +
  # uses a clean theme
  theme_minimal()
```



DO Bins Across Sites

```
# create new object to store medians of median do levels
median_fig2 <- aq_ins_clean_fig2 |>
  group_by(site_full) |>
  summarize(median_do = median(med_do, na.rm = TRUE))

# assign quartiles for medians for each site
median_fig2$quartile <- cut(median_fig2$median_do,
                   breaks = quantile(median_fig2$median_do,
                                     probs = seq(0, 1, 0.25)),
                   include.lowest = TRUE,
                   labels = c("Q1","Q2","Q3","Q4"))

# boundaries of quartiles
q <- quantile(median_fig2$median_do, probs = c(0.25, 0.5, 0.75))

# distribution plot of DO levels across sites
ggplot(median_fig2, aes(x = median_do)) +
  # density curve
  geom_density(fill = "lightblue",
               alpha = 0.4,
               color = "blue",
               linewidth = 1.2) +
  # vertical quartile lines
  geom_vline(xintercept = q,
             linetype = "dashed",
             color = "gray40") +
  # site labels
  geom_text(aes(y = 0.01,
                label = str_wrap(site_full, width = 10),
                color = quartile),
            hjust = 0.5,
            size = 3) +
  # remove space between bottom of plot and y-axis, set y-axis limits
  scale_y_continuous(expand = c(0,0),
                     limits = c(0, 0.15)) +
  # axes labels and title
  labs(x = "Median Dissolved Oxygen (mg/L)",
       y = "Density",
       title = "Distribution of Median Dissolved Oxygen Aross Sites",
```

```r
      color = "Quartile") +
  # theme
  theme_bw()
```

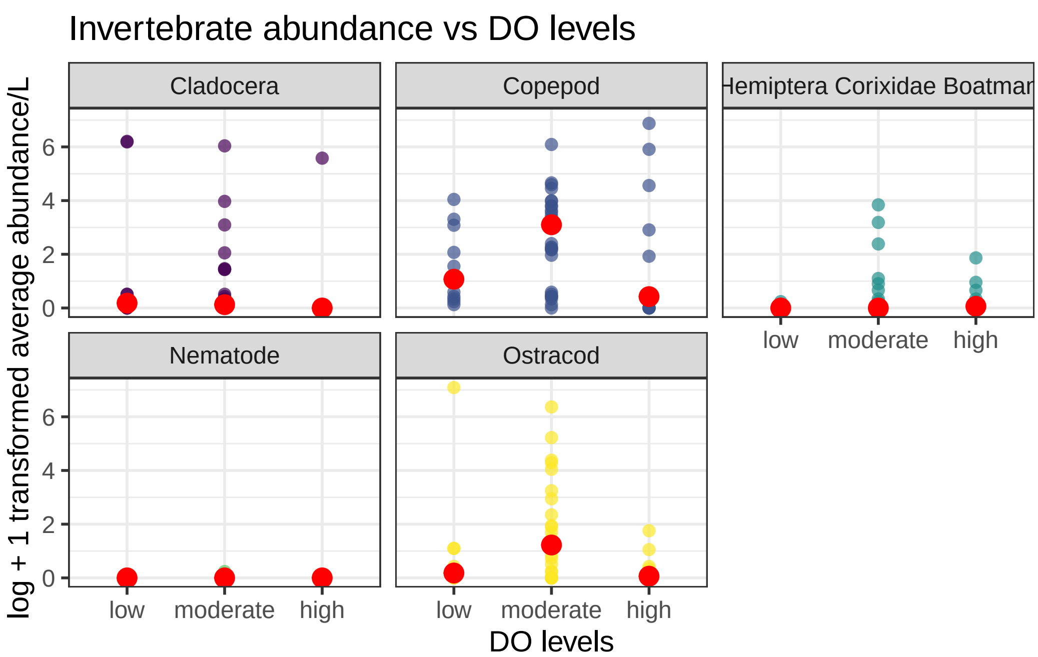


BINS Low DO: Phelps Road and North of Phelps Bridge Moderate DO: South of Venoco Bridge, South of Phelps Bridge, Devereux Creek High DO: North of Venoco Bridge, Main Channel

Invertebrate Abundance in DO Bins

```r
aq_ins_clean_bins <- aq_ins_clean |>
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
```

```r
                              site_full == "North of Phelps Bridge" ~ "low",
                              site_full == "South of Venoco Bridge" ~ "moderate",
                              site_full == "South of Phelps Bridge" ~ "moderate",
                              site_full =="Devereux Creek" ~ "moderate",
                              site_full =="North of Venoco Bridge" ~ "high",
                              site_full == "Main Channel" ~ "high")) |>
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))


# base layer: ggplot
ins_bins_plot <- ggplot(data = aq_ins_clean_bins,
                        mapping = aes(x = bins, # x-axis: DO bins
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
             size = 2, # size is 2
             alpha = 0.7) + # opacity is 0.7
  # calculate median for each bin for each taxon
  stat_summary(fun = median,
               geom = "point",
               color = "red",
               size = 3) +
  # separate panels for each taxon
  facet_wrap(~ taxon) +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "DO levels", # relabel x-axis
       y = "log + 1 transformed average abundance/L", # relabel y-axis
       # creates descriptive title
       title = "Invertebrate abundance vs DO levels") +
  # uses non-default theme
  theme_bw() +
  # remove redundant legend
  theme(legend.position = "none")

# shows plot
ins_bins_plot
```




```
# Kruskal-Wallis test between abundance and bins for Copepods
kruskal.test(log_ave_lit ~ bins,
             data = aq_ins_clean_bins,
             subset = taxon == "Copepod") # Kruskal-Wallis chi-squared = 4.405, df = 2, p-val
```

```

    Kruskal-Wallis rank sum test

data:  log_ave_lit by bins
Kruskal-Wallis chi-squared = 4.405, df = 2, p-value = 0.1105
```

```
                                          # no significant difference

# Kruskal-Wallis test between abundance and bins for Ostracods
kruskal.test(log_ave_lit ~ bins,
             data = aq_ins_clean_bins,
             subset = taxon == "Ostracod") # Kruskal-Wallis chi-squared = 6.7235, df = 2, p-v
```

```

    Kruskal-Wallis rank sum test

data:  log_ave_lit by bins
```

```
Kruskal-Wallis chi-squared = 6.7235, df = 2, p-value = 0.03467
```

```
                                            # significant difference of abundance btwn DO leve
```

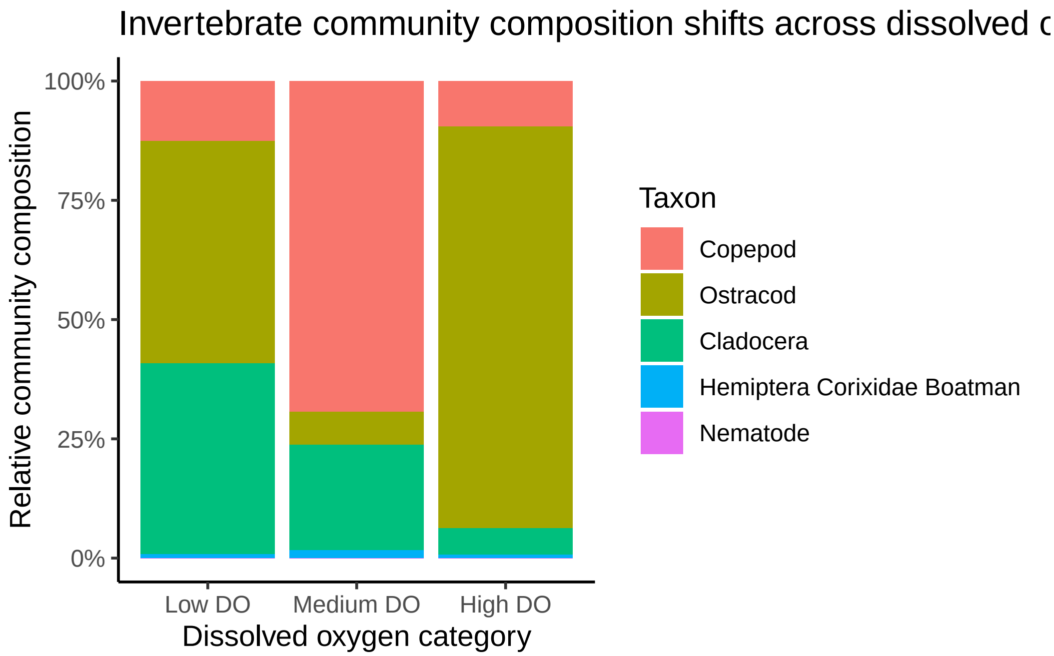
```
# Pairwise test between abundance and individual bins for Ostracods
bins_ostracods <- aq_ins_clean_bins |>
  filter(taxon == "Ostracod")
pairwise.wilcox.test(
  bins_ostracods$log_ave_lit,
  bins_ostracods$bins,
  p.adjust.method = "bonferroni") # p = 0.051 for comparison btwn moderate and high DO level
```

```
    Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data:  bins_ostracods$log_ave_lit and bins_ostracods$bins

         low   moderate
moderate 0.387 -
high     1.000 0.051

P value adjustment method: bonferroni
```

From this figure and the medians, it looks like Copepods and Ostracods prefer moderate levels of DO. The others don't appear to have a preference, but nematode abundance is especially low throughout so there might just not be enough to tell for that taxon. Only Ostracod abundance medians for each DO level bin significantly differ according to a Kruskal-Wallis test. A pairwise comparison using Wilcoxon rank sum test found that abundance compared between moderate and high levels of DO for Ostracods has a p-value of 0.051. Otherwise, pairwise comparisons are not significant (p>0.1).

```
# Made a copy of the composition plot but changed the bins to match the 4 quartiles of the di
composition_plot_1 <- DO_taxa |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.25, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.75, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    ),
    do_category = factor(do_category,
```

```r
                         levels = c("Low DO",
                                    "Medium DO",
                                    "High DO"))
  ) |>
  group_by(do_category, taxon) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  ggplot(aes(x = do_category,
             y = total_abundance,
             fill = taxon)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    x = "Dissolved oxygen category",
    y = "Relative community composition",
    fill = "Taxon",
    title = "Invertebrate community composition shifts across dissolved oxygen levels"
  ) +
  theme_classic()

# display plot
composition_plot_1
```




```r
# running chi-square tests to compare proportional abundance across DO levels
copepods_matrix <- matrix(c(160.133333,1112.133333, 2155.200000,949.866667, 17.466667,165.066
                          dimnames = list(Oxygen = c("Low", "Moderate", "High"),
                                          Taxa = c("Copepod", "Other")))
view(copepods_matrix)
chisq.test(copepods_matrix)
```

```
    Pearson's Chi-squared test

data:  copepods_matrix
X-squared = 1297.8, df = 2, p-value < 2.2e-16
```

```r
# X-squared = 1297.8, df = 2, p-value < 2.2e-16
# Proportion of Copepod abundance changes across DO levels.
pairwise_copepods <- pairwise.prop.test(copepods_matrix, p.adjust.method = "bonferroni")
pairwise_copepods
```

```
    Pairwise comparisons using Pairwise comparison of proportions

data:  copepods_matrix
```

```
         Low    Moderate
Moderate <2e-16 -
High     0.89   <2e-16

P value adjustment method: bonferroni
```

```r
# Significant difference between Low & High, Moderate & High

ostracods_matrix <- matrix(c(592.666667,679.599963, 211.466667,2893.600033, 153.733333,28.800
                      dimnames = list(Oxygen = c("Low", "Moderate", "High"),
                                      Taxa = c("Copepod", "Other")))
view(ostracods_matrix)
chisq.test(ostracods_matrix)
```

```
    Pearson's Chi-squared test

data:  ostracods_matrix
X-squared = 1318.3, df = 2, p-value < 2.2e-16
```

```r
# X-squared = 1318.3, df = 2, p-value < 2.2e-16
# Proportion of Ostracod abundance changes across DO levels.
pairwise_ostracods <- pairwise.prop.test(ostracods_matrix, p.adjust.method = "bonferroni")
pairwise_ostracods
```

```
    Pairwise comparisons using Pairwise comparison of proportions

data:  ostracods_matrix

         Low    Moderate
Moderate <2e-16 -
High     <2e-16 <2e-16

P value adjustment method: bonferroni
```

```r
# Significant difference between all pairs of DO levels.

cladocera_matrix <- matrix(c(509.333333,762.933333, 688.400000,2417.06667, 10.000000,155.0666
                      dimnames = list(Oxygen = c("Low", "Moderate", "High"),
```

```r
                                      Taxa = c("Copepod", "Other")))
view(cladocera_matrix)
chisq.test(cladocera_matrix)
```

```
    Pearson's Chi-squared test

data:  cladocera_matrix
X-squared = 184.58, df = 2, p-value < 2.2e-16
```

```r
# X-squared = 184.58, df = 2, p-value < 2.2e-16
# Proportion of Cladocera abundance changes across DO levels.
pairwise_cladocera <- pairwise.prop.test(cladocera_matrix, p.adjust.method = "bonferroni")
pairwise_cladocera
```

```
    Pairwise comparisons using Pairwise comparison of proportions

data:  cladocera_matrix

         Low     Moderate
Moderate < 2e-16 -
High     < 2e-16 4.2e-06

P value adjustment method: bonferroni
```

```r
# Significant difference between all pairs of DO levels.

boatman_matrix <- matrix(c(10.133333,1262.133333, 50.000000,3055.466667, 1.333333,181.2), nro
                      dimnames = list(Oxygen = c("Low", "Moderate", "High"),
                                      Taxa = c("Copepod", "Other")))
view(boatman_matrix)
chisq.test(boatman_matrix)
```

```
    Pearson's Chi-squared test

data:  boatman_matrix
X-squared = 5.0378, df = 2, p-value = 0.08055
```

```
# X-squared = 5.0378, df = 2, p-value = 0.08055
# Proportion of Boatman abundance does not change across DO levels.
```

When comparing medians of taxon abundance across DO bins, only Ostracod abundance is significantly different between the moderate and high DO bins. When comparing proportions of taxon abundnace across DO bins, the proportions of Copepods, Ostracods, and Cladocera significantly differ across the different DO bins. For Copepods, the differences exist between Low & High and Moderate & High DO levels. For Ostracods and Cladocera, the differences exist between all pairings of DO levels.

Chi-square

```
# create abundance table for chi-square test
chi_square_table <- aq_ins_clean_bins |>
  filter(taxon %in% focus_taxa) |>
  group_by(bins, taxon) |>
  summarise(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  pivot_wider(
    names_from = taxon,
    values_from = total_abundance,
    values_fill = 0
  ) |>
  column_to_rownames("bins") |>
  as.matrix()

# show table used in chi-square test
chi_square_table
```

```
         Cladocera   Copepod Hemiptera Corixidae Boatman  Nematode    Ostracod
low       980.2667  115.8667                   0.2666667 0.1333333 1202.933333
moderate  510.6667 1170.4000                  83.8666667 0.2666667 1057.866667
high      264.2667 1453.4667                   8.8000000 0.0000000    7.866667
```

```
# chi-square test for differences in taxon composition across DO bins
chi_square_test <- chisq.test(chi_square_table)
```

```r
# show chi-square result
chi_square_test
```

```
	Pearson's Chi-squared test

data:  chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16
```

A chi-square test was used to compare whether the proportional composition of the focus taxa differed across low, moderate, and high DO bins. The chi-square test suggests that taxon composition differs strongly across DO bins: the relative abundance of Cladocera, Copepod, Boatman, Nematode, and Ostracod is not evenly distributed across low, moderate, and high DO groups, $X^2 = 2859.2$, df = 8, $p < 2.2e-16$